

Final Examination, Autumn-2024

Course Title: Chemistry

Course Code: CHEM-2301

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Group-A

1. a) Define electrolytes. Differentiate between ionization and dissociation.

Ans: Electrolytes are substances that conduct electricity in their molten state or in aqueous solution by producing ions.

Ionization is the process where neutral molecules form ions, typically in covalent compounds (e.g., $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ in water), while **Dissociation** refers to the separation of existing ions in an ionic compound when dissolved (e.g., $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$ in water).

1. b) Discuss the Faraday's Laws of electrolysis with relevant equations.

Ans: First Law: The mass of a substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed.

$m \propto Q \Rightarrow m = (E \times Q) / 96500$, where E = equivalent weight, Q = total charge in coulombs.

Second Law: When the same amount of electricity is passed through different electrolytes, the masses of substances deposited are proportional to their equivalent weights.

$$m_1/m_2 = E_1/E_2$$

1. c) A Solution of AgNO_3 Containing 12.14 g of Silver in 50 ml of Solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

Ans: Initial mass of Ag = 12.14 g

Final mass of Ag = 11.55 g

Loss of Ag^+ from solution = $12.14 - 11.55 = 0.59$ g

Mass deposited at cathode = 1.25 g

$$\text{Transport number of } \text{Ag}^+ (t_+): t_+ = \frac{\text{Ag deposited}}{\text{Ag deposited} + \text{Ag lost}} = \frac{1.25}{1.25 + 0.59} \approx 0.679$$

Transport number of NO_3^- (t_-): $t_- = 1 - t_+ = 1 - 0.679 = 0.321$

2. a) Discuss Henry's law for the solubility of a gas in a solvent.

Ans: Henry's law states that the mass of a gas dissolved in a given volume of solvent at constant temperature is directly proportional to the pressure of the gas in equilibrium with the solvent. Mathematically, it is expressed as:

$$w \propto P \text{ or } w = kP,$$

where w is the mass of gas dissolved, P is the pressure, and k is Henry's constant depending on the nature of the gas, solvent, and temperature.

Another form of the law states that the volume of gas dissolved at the pressure used is independent of the pressure itself. Henry's law is valid for **dilute solutions of non-reacting gases**, and when a mixture of gases is in contact with a liquid, the **partial pressure** of each gas determines its solubility.

Application:

It is applied in making **carbonated drinks** like soda and beer, where CO_2 is dissolved under high pressure (~ 4 atm). On opening the bottle, the **pressure drops (~ 0.05 atm)** and CO_2 escapes as bubbles.

Limitations:

Henry's law holds well for gases of low solubility, under **moderate pressure and temperature**, and when the gas does not chemically react with the solvent. It strictly applies when the **molecular species remain the same** in both gas and liquid phases.

2. b) Distinguish between Ideal and Real solution.

Ans:

Point of Difference	Ideal Solution	Real (Non-Ideal) Solution
Raoult's Law	Obeys Raoult's law at all temperatures and pressures.	Deviates from Raoult's law.
Enthalpy of Mixing (ΔH_{mix})	$\Delta H_{\text{mix}} = 0$ (no heat absorbed or evolved during mixing).	$\Delta H_{\text{mix}} \neq 0$ (heat may be absorbed or evolved).
Volume Change (ΔV_{mix})	$\Delta V_{\text{mix}} = 0$ (no volume change occurs on mixing).	$\Delta V_{\text{mix}} \neq 0$ (volume change occurs during mixing).

Intermolecular Forces	Intermolecular forces between A–A, B–B, and A–B are identical.	Intermolecular forces between solute and solvent are different.
Entropy of Mixing (ΔS_{mix})	$\Delta S_{\text{mix}} = -R \sum x_i \ln x_i$ (follows ideal entropy expression).	Entropy may deviate from ideal behavior.
Examples	Hexane + Heptane, Benzene + Toluene	Water + Ethanol, HCl + Water
Reasons for Non-Ideality	—	High concentration, association/dissociation, pressure difference, compound formation

2. c) What is colligative property. Explain Raoult's law for the relative lowering of vapour pressure with its mathematical expression.

Ans: Colligative properties are physical properties of solutions that depend only on the number of solute particles present, not their chemical nature. These include:

- Relative lowering of vapor pressure
- Elevation of boiling point
- Depression of freezing point
- Osmotic pressure

They arise due to the presence of solute particles in a solvent, affecting the escaping tendency of solvent molecules.

Raoult's law states that the **vapor pressure of a solvent is lowered** when a non-volatile solute is added to it, and the **relative lowering of vapor pressure** is equal to the mole fraction of the solute.

$$\frac{P_0 - P}{P_0} = X_{\text{solute}}$$

Where:

- P_0 = vapor pressure of pure solvent
- P = vapor pressure of solution
- X_{solute} = mole fraction of solute

If n_2 is moles of solute and n_1 is moles of solvent:

$$\frac{P_0 - P}{P_0} = \frac{n_2}{n_1 + n_2}$$

For dilute solutions, $n_2 \ll n_1$, so:

$$\frac{P_0 - P}{P_0} \approx \frac{n_2}{n_1}$$

This law forms the basis for other colligative properties and helps in determining molar mass of solutes.

2. d) Describe the relation between elevation of boiling point and lowering of vapour pressure and deduce a mathematical expression for the determination of molecular mass from elevation of boiling point.

Ans: When a **non-volatile solute** is added to a solvent, the **vapour pressure of the solution decreases**. As a result, the solution requires a **higher temperature to boil**, leading to **elevation of boiling point**. This is a **colligative property**, depending only on the number of solute particles, not their nature.

Mathematical Expression for Determining Molecular Mass:

$$\Delta T_b = K_b \cdot m = K_b \cdot \frac{w_2 \cdot 1000}{M_2 \cdot w_1}$$

Solving for molar mass:

$$M_2 = \frac{K_b \cdot 1000 \cdot w_2}{\Delta T_b \cdot w_1}$$

Where:

- ΔT_b = Elevation in boiling point
- K_b = Molal elevation constant
- m = Molality of the solution
- w_2 = mass of solute (g)
- w_1 = mass of solvent (g)
- M_2 = molar mass of solute (g/mol)

OR

2. a) Define conductance and molar conductance.

Ans: Conductance is the ability of a solution to conduct electric current. It is the reciprocal of resistance and is denoted by **G**. Mathematically,

$$G = \frac{1}{R}$$

Where **R** is the resistance in ohms (Ω). The unit of conductance is **siemens (S)** or **ohm⁻¹**.

Molar conductance (Λ_m) is the conductance of all the ions produced by one mole of an electrolyte in a given solution. It depends on both the concentration and the mobility of ions. It is given by:

$$\Lambda_m = \kappa \times \frac{1000}{C}$$

Where,

- Λ_m = molar conductance ($S \cdot cm^2 \cdot mol^{-1}$)
- κ (**kappa**) = specific conductance or conductivity ($S \cdot cm^{-1}$)
- **C** = molar concentration ($mol \cdot L^{-1}$)

2. b) Classify the following electrolytes as strong and weak electrolytes. $HClO_4$, H_2SO_3 , NH_4OH , HNO_3 .

Ans: $HClO_4$ (Perchloric acid) – Strong electrolyte

H_2SO_3 (Sulfurous acid) – Weak electrolyte

NH_4OH (Ammonium hydroxide) – Weak electrolyte

HNO_3 (Nitric acid) – Strong electrolyte

2. c) What is transport number? Prove that, $t_- = 1 / 1+r$, where, t_- is the transport number of the anion and r is the speed ratio of the ions.

Ans: Transport number (t) is the fraction of the total current carried by a particular ion in an electrolyte solution.

Let:

- t_+ = transport number of cation
- t_- = transport number of anion
- u_+ = speed (or mobility) of cation
- u_- = speed (or mobility) of anion
- $r = u_+ / u_-$ (speed ratio of cation to anion)

We know that:

$$t_+ = \frac{u_+}{u_+ + u_-}, t_- = \frac{u_-}{u_+ + u_-}$$

Dividing numerator and denominator of t_- by u_- , we get:

$$t_- = \frac{1}{\frac{u_+}{u_-} + 1} = \frac{1}{r + 1}$$

Thus,

$$t_- = \frac{1}{1 + r}$$

2. d) The resistance of decinormal solution of a salt occupying a volume between two-platinum electrodes 1.80 cm apart and 5.4 cm² in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.

Ans: Given:

- **Resistance (R)** = 32 Ω
- **Distance between electrodes (l)** = 1.80 cm
- **Area of electrodes (A)** = 5.4 cm²
- **Normality (N)** = 0.1 N (decinormal solution)

Step 1: Calculate conductivity (κ)

$$\kappa = \frac{l}{RA} = \frac{1.80}{32 \times 5.4} \approx 0.01042 \text{ S}\cdot\text{cm}^{-1}$$

Step 2: Calculate equivalent conductance (Λ_{eq})

$$\Lambda_{eq} = \kappa \times \frac{1000}{N} = 0.01042 \times \frac{1000}{0.1} = 104.2 \text{ S}\cdot\text{cm}^2 \cdot \text{eq}^{-1}$$

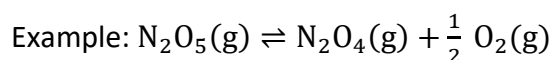
Group-B

3. a) Define Zero order reaction and Pseudo-unimolecular reaction & explain briefly with suitable examples.

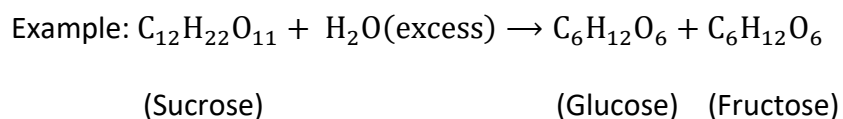
Ans: The reaction whose rate is directly proportional to the first power of concentration of reactant is called first order reaction.

According to the rate law, first order reaction rate is:

$$\frac{-dc}{dt} = \frac{dx}{dt} = kc^1 = kc$$



It is a reaction which seems to be second order, but actually it is first order reaction. In this case, one reactant remains in excess and the reaction rate depends on the concentration of the other reactant — is called pseudo unimolecular reaction.

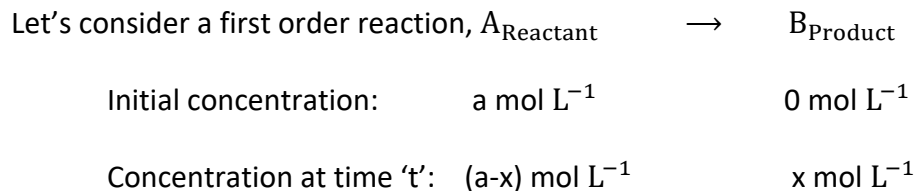


The rate of this pseudo unimolecular reaction is:

$$\text{Rate} = k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]$$

3. b) Define first order reaction. Prove that $K = 2.303 \log \frac{a}{a-x}$ Where symbols have usual meanings.

Ans: A first order reaction is a chemical reaction whose rate is directly proportional to the concentration of a single reactant raised to the first power.



According to the law of mass action, the rate of the first order reaction:

$$\frac{dx}{dt} = k(a - x)$$

$$\Rightarrow \frac{dx}{a-x} = kdt \dots \dots (1)$$

Integrating equation (1) in both sides,

$$\int \frac{dx}{a-x} = \int kdt$$

$$\Rightarrow -\ln(a-x) = kt + c \dots \dots (2)$$

When $t=0$ sec, then $x=0 \text{ mol L}^{-1}$

Putting the values of 't' and 'x' in equation (2),

$$-\ln(a-0) = k \cdot 0 + c$$

$$\therefore c = -\ln a$$

Now from equation (2),

$$-\ln(a-x) = kt - \ln a$$

$$\Rightarrow \ln a - \ln(a-x) = kt$$

$$\Rightarrow \ln a - \ln(a-x) = kt$$

$$\Rightarrow \ln \frac{a}{a-x} = kt$$

$$\Rightarrow k = \frac{1}{t} \ln \frac{a}{a-x}$$

$$\therefore k = \frac{2.303}{t} \log \frac{a}{a-x}$$

Where:

- K = Rate Constant
- a = Initial concentration of reactant A
- x = Increased concentration of product B or decreased concentration of reactant A
- t = Time of which reaction proceeds

3. c) Differentiate between order of reaction and molecularity.

Ans:

Aspect	Order of Reaction	Molecularity
Definition	Sum of powers of concentration terms in rate law	Number of molecules involved in elementary step
Type	Experimental (determined from data)	Theoretical (based on mechanism)
Can be Fractional?	Yes, order can be fractional or zero	No, molecularity is always a whole number
Applies to	Overall reaction (can be complex)	Only to elementary reactions
Example	Rate = $k[A]^1[B]^{\frac{1}{2}} \Rightarrow$ order = 1.5	Reaction: $A + B \rightarrow$ Products $\Rightarrow$ molecularity = 2

4. a) Chemical equilibrium is a dynamic equilibrium - explain it.

Ans: Chemical equilibrium is a dynamic state where the rate of the forward reaction equals the rate of the backward reaction. Although the concentrations of reactants and products remain constant, both reactions continue to occur at the molecular level.

4. b) Discuss the law of mass action. Express the equilibrium constant for the following reaction:



Ans: The law of mass action states that the rate of a chemical reaction is proportional to the product of the active masses (concentrations) of the reactants. For the reaction $A + B \rightleftharpoons C + D$, the equilibrium constant (K_c) is given by:

$$K_c = \frac{[C][D]}{[A][B]}$$

4. c) Prove that $K_p = K_c \times (RT)^{\Delta n}$

Ans: Let us consider a general gas-phase reaction: $aA + bB \rightleftharpoons cC + dD$

- $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$, using concentrations.
- $K_p = \frac{(P_C)^c(P_D)^d}{(P_A)^a(P_B)^b}$, using partial pressures.
- From the ideal gas law, $P_X = [X]RT$.

Substitute $P_X = [X]RT$ into K_p :

$$K_p = \frac{([C]RT)^c([D]RT)^d}{([A]RT)^a([B]RT)^b} = \frac{[C]^c[D]^d}{[A]^a[B]^b} \cdot (RT)^{(c+d)-(a+b)}$$

Since $\frac{[C]^c[D]^d}{[A]^a[B]^b} = K_c$ and $(c + d) - (a + b) = \Delta n$:

$$K_p = K_c \times (RT)^{\Delta n}$$

5. a) What are colloids? Give some examples of colloids.

Ans: Colloids are mixtures in which one substance is dispersed in another in the form of tiny particles. These particles do not settle and cannot be filtered. Examples include milk, fog, and gelatin.

5. b) What is an emulsion? Discuss the role of emulsifier in a solution.

Ans: An emulsion is a type of colloid where two immiscible liquids are mixed, such as oil and water. An emulsifier helps stabilize the emulsion by reducing surface tension and preventing the liquids from separating. Examples: soap in water-oil mixture.

5. c) Illustrate Freundlich adsorption isotherm with relevant log plot.

Ans: Freundlich proposed an empirical relation in the form of a mathematical equation.

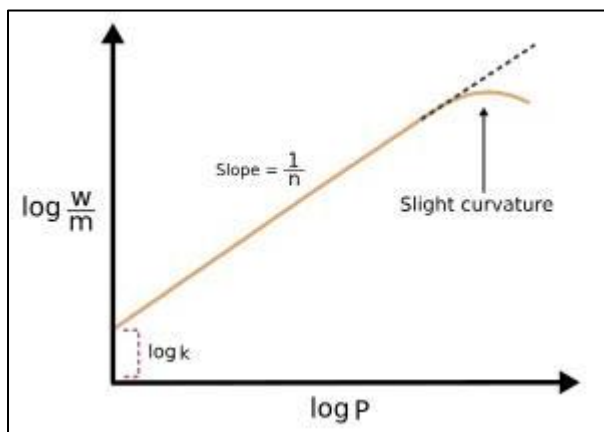
$$\frac{w}{m} = kP^{1/n}$$

where w is the mass of the gas adsorbed on a mass m of adsorbent at a pressure P ; k and n are constants depending on the nature of the gas and the adsorbent and on temperature. This relation is generally represented in the form of a curve obtained by plotting the mass of the gas adsorbed per unit mass of adsorbent (w/m) against equilibrium pressure. Freundlich isotherm is not applicable at high pressures. Taking logarithms on both sides of Freundlich equation, we have

$$\log \frac{w}{m} = \log k + \frac{1}{n} \log P$$

This is equation for a straight line. Thus a plot of $\log (w/m)$ against $\log P$ should be a straight line with slope $1/n$ and intercept $\log k$. However, it is actually found that the plots were straight

lines at low pressures, while at higher pressure they showed a slight curvature, especially at low temperature. This indicated that Freundlich equation is approximate and does not apply to adsorption of gases by solids at higher pressures.



5. d) What is Langmuir adsorption isotherm? Write the important assumptions in Langmuir adsorption isotherm.

Ans: Langmuir adsorption isotherm is a model that describes the adsorption of gas molecules onto a solid surface, forming a monolayer.

Langmuir made the following assumptions:

- (1) The layer of the gas adsorbed on the solid adsorbent is one-molecule thick.
- (2) The adsorbed layer is uniform all over the adsorbent.
- (3) There is no interaction between the adjacent adsorbed molecules.

OR

5. a) Differentiate between homogeneous equilibrium and heterogeneous equilibrium.

Ans: In homogeneous equilibrium, all reactants and products are in the same phase (e.g., gases or solutions), while in heterogeneous equilibrium, they exist in different phases (e.g., solids and gases).

5. b) Discuss the characteristics of chemical equilibrium.

Ans: There are some characteristics of chemical equilibrium. They are –

1. **Reversibility of Reaction:** Chemical equilibrium is related only to reversible reactions. It is not applicable to irreversible reaction.
2. **Stability of equilibrium:** If the external conditions (pressure, temperature and concentration) are not changed, the system will stay forever.
3. **Easy approachability from both sides:** The equilibrium can be attained from both directions.
4. **Closed system:** It is only attained in closed system.
5. **Ineffectiveness of catalysts:** The catalysts have no effect on chemical equilibrium.

5. c) Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N₂, 2.10 moles of H₂, and 0.298 mole of NH₃, are found to be present. Find the value of K_c, for the reaction N₂(g)+3H₂(g) ⇌ 2NH₃(g) at 500°C.

Ans: Given: N₂ = 3.00 mol, H₂ = 2.10 mol, NH₃ = 0.298 mol, V = 5 L

Reaction: N₂(g) + 3H₂(g) ⇌ 2NH₃(g)

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$$K_c = \frac{(0.298/5)^2}{((3.00/5) \times (2.10/5)^3)} = \frac{(0.0596)^2}{(0.6 \times 0.074)}$$

$$K_c \approx 0.00355 / 0.0444 \approx \mathbf{0.0799 \text{ mol}^{-2}\text{L}^2}$$

5. d) What are the conditions (temperature, pressure and catalyst) for maximum production of Ammonia by Heber process in industry.

Ans: Conditions for maximum production of ammonia by Haber process in industry:

- **Temperature:** Around **450°C** — a moderate temperature that balances the exothermic nature of the reaction and ensures a reasonable reaction rate.
- **Pressure:** About **200 atmospheres (15–25 MPa)** — high pressure favors the formation of ammonia as it results in fewer gas molecules.
- **Catalyst:** **Iron-based catalyst** promoted with **aluminum oxide (Al₂O₃)** and **potassium hydroxide (KOH)** to increase efficiency and reaction rate.